

DevOps From IBM Perspective

Accelerating innovation through AI-driven methodologies and high-performance automation ecosystems.

Prepared By:

Ishita

SAP ID:

500119435

| Redefining DevOps



A Methodology, Not a Tool

DevOps is a set of practices that combines software development (Dev) and IT operations (Ops). It aims to shorten the systems development life cycle and provide continuous delivery with high software quality.



The IBM Strategic View

From IBM's perspective, it is the foundation of [Rapid Application Development \(RAD\)](#), designed to synchronize and automate tasks across the entire organizational lifecycle.

| The Need for Transformation

60%

Faster Time to Market

By leveraging tools that promote radical coordination and agile collaboration across silos.

-  **Streamlined Lifecycles:** Integrating small code changes that are tested and deployed rapidly.
-  **Risk Assessment:** Enabling teams to focus on risks by assessing true impact and contextual relevance.
-  **Silo Elimination:** Removing the barriers between creation and maintenance teams.

| The Pillars of IBM's Framework



Culture

Collaboration and
shared responsibility
across teams.



Process

Streamlined
workflow through
lean and agile
automation.



Tools

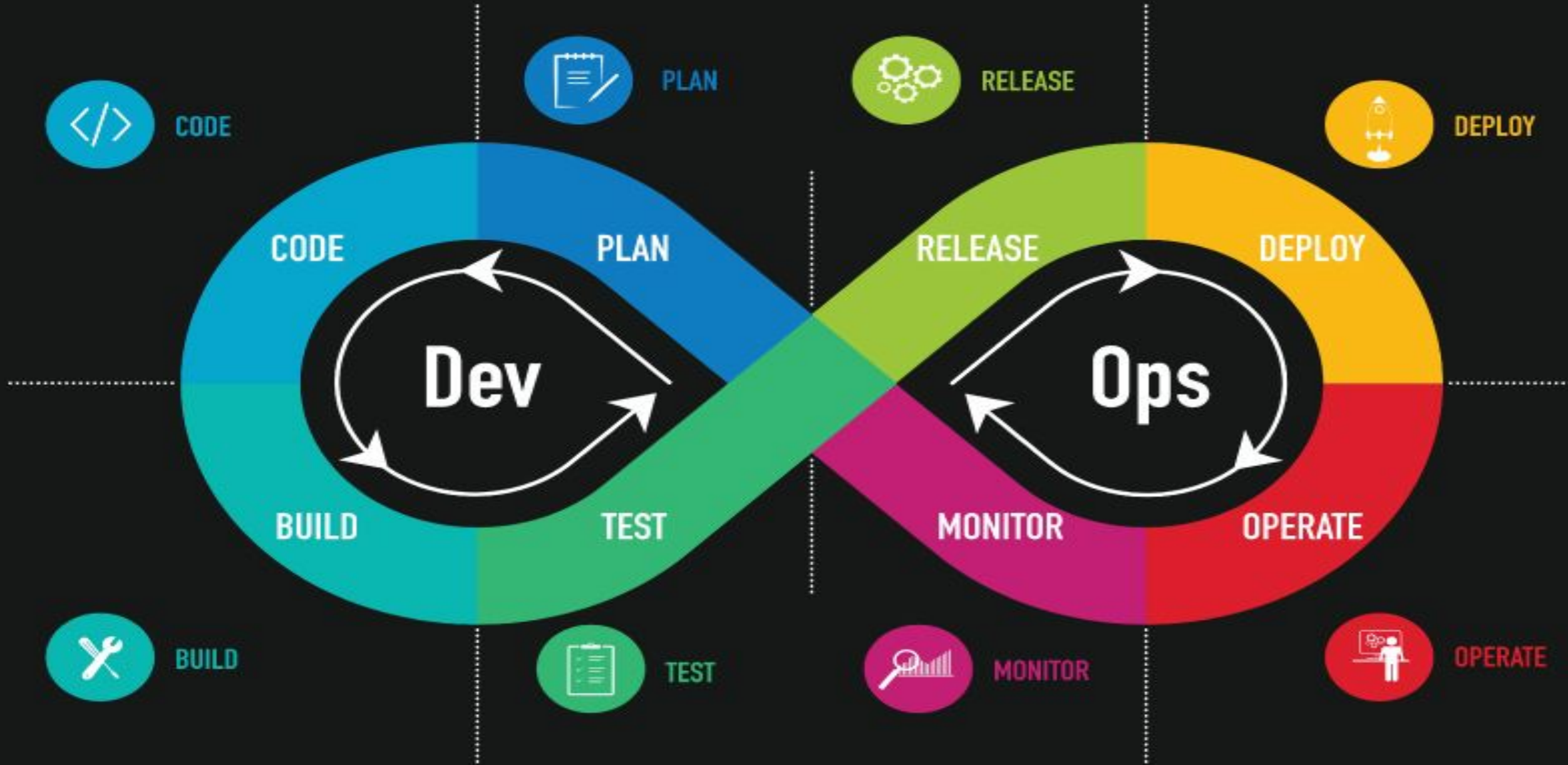
Strategic technology
stack for continuous
delivery.



Analysis

Tracking
performance and
continuous
monitoring.

The DevOps Lifecycle



| Lifecycle: Plan to Test

Phase	Core Objective	Key Activities
Planning	Define Backlog	User feedback analysis, scoping features & fixes.
Coding	Feature Dev	Test-driven development (TDD), pair programming.
Building	Integration	Automated merging, compilation into binary repos.
Testing	Quality Assurance	Automated unit & integration tests, shift-left testing.

| Lifecycle: Release to Monitor

Phase	Core Objective	Key Activities
Release	Validation	Compliance checks, security audits, final staging.
Deploy	Go-Live	Infrastructure as Code (IaC), gradual rollouts.
Operate	Reliability	Auto-scaling, performance optimization, uptime mgt.
Monitor	Insights	Real-time telemetry, user feedback loops.

| DevOps + AI

- 🔧 **Predictive Scaling:** AI predicts and processes bottlenecks before they occur.
- 🛡️ **Dynamic Security:** Real-time detection of data breaches and supply chain attacks.
- 🐞 **Anomaly Detection:** Identifying hidden bugs and flagging performance fluctuations.
- 🔧 **Automated Remediation:** Self-healing systems based on historical data.



| Strategic Value for Enterprise

 **Collaboration:** Synchronizing Dev & Ops into a single cohesive unit.

 **Speed:** Drastically faster release cycles via CI/CD pipelines.

 **Stability:** Continuous testing ensures system reliability.

 **Scaling:** Automating deployment to scale with global demand.

 **Security:** Compliance as code reduces vulnerabilities.

 **Efficiency:** Reducing remediation costs through early detection.

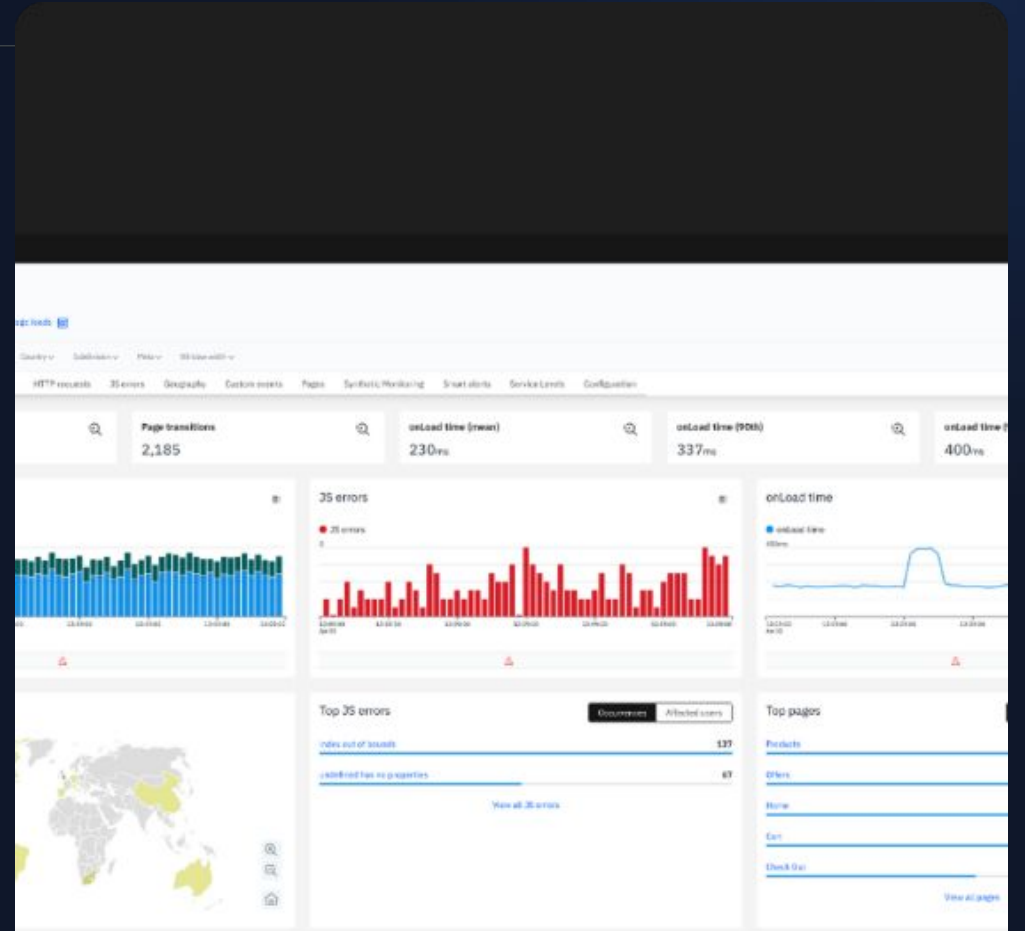
IBM's Tooling Ecosystem

IBM Instana

Proactive monitoring and observability for CI/CD with early-stage issue detection.

IBM Concert

Securing source-to-runtime via vulnerability management integrated into IDEs.



References & Insights

Source	Key Takeaway
The 2026 Guide to DevOps	Integrating AI-driven automation to accelerate delivery.
IBM DevOps Solutions	Modern DevOps uses observability and security to reduce risk.
Industry Standards (CALMS)	Emphasizing Culture, Automation, Lean, and Measurement.

Debug Your Doubts

Q1: How does IBM's approach differ?

IBM focuses on AI-driven practices and predictive monitoring for enterprise-scale scalability.

Q2: Why is Shift-Left Testing important?

It ensures early bug detection, significantly reducing the cost and effort of remediation.

**Thank
You!**